Steffen **Plunder**

Mathematician · Developer · Interdisciplinary Researcher

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Work Experience

Research Scientist, ASHBi, Kyoto University, Japan *02/2023 – present*

- Built GPU-accelerated simulations for digit growth and analyzed 3D image data (meshing, cell tracks, structural measures).
- Published in high impact journals Nature Physics, Nature Materials, and iScience.
- Initiated cross-department networking events to connect researchers across biology and mathematics. Co-supervised three students, co-organized international workshops.

Graduate Researcher, University of Vienna, Austria *10/2019 – 01/2023*

- Proved that a widely-used game engine technique (Position-based Dynamics) is scientifically accurate.
- Achieved 300× speedup in simulations through algorithm design and efficient implementation.
- Published two Nature Communications papers and 4 further manuscripts.

Research Assistant, TU Kaiserslautern, Germany *12/2018 – 10/2019*

- Organized calculus classes and exams. Research in Differential-Algebraic Systems group leading to 2 publications.

Programming Assistant, Fraunhofer Institute ITWM, Kaiserslautern, Germany *06/2013 – 06/2016*

- C++ programming for image analysis software and user interfaces (Qt).

Education

German grading: 1 = excellent, 4 = passing, 6 = fail.

Ph.D. in Mathematics, University of Vienna, Austria Grade: 1.0 10/2019 – 01/2023

M.Sc. in Mathematics, TU Kaiserslautern, Germany Grade: 1.1 10/2016 – 09/2018

B.Sc. in Mathematics, TU Kaiserslautern, Germany Grade: 1.1 10/2013 – 09/2016

Awards and Grants

Kakenhi: Early-Career Scientists (PI) ≈ 12,000€ 04/2024 – 03/2027
Building and analyzing multicellular simulations in NVIDIA Omniverse

ASHBi Fusion Grant (Co-PI) ≈ 30,000€ 04/2023 – 03/2025
Co-led interdisciplinary biology–mathematics collaboration across two research groups

Further Scholarships: EMBO Scientific Fellowship (2021), DAAD-PROMOS Scholarship (2018), Main Award DMV Students Conference (2017), Felix-Klein Scholarship (2016–2018), Deutschlandstipendium (2014–2016)

Skills

Technical: GPU-accelerated simulation, agent-based modeling, numerical solvers, performance optimization

Programming: Julia, Python, C++, JavaScript/TypeScript, MATLAB, Lua

Frameworks: SciML, numpy/scipy, numba, pytorch, FEniCSx, CUDA, Omniverse, react/nextjs, Git/GitHub, Linux

AI Tools: GitHub Copilot (since 2023), Claude Code / Codex (1 yr daily use, tutorial instructor), Superpowers, Amplifier, Vibe/Mistral, Ollama

Languages: German (Native), English (Fluent), Chinese (Basic A1-A2)

Personal Note

I work across computer science, biology, and computational mathematics, coding with AI tools daily and introducing many colleagues to modern AI-assisted workflows. My aim: create real value, connect with people, and act with emotional intelligence and brave ideas.